

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 3: Aerodrome Operating Minima (AOM) & LVO

Syllabus Source: CAE Oxford (pp. 77-116) | Pass Target: >= 90% AviationExam

1. ALL-WEATHER PRECISION APPROACH CATEGORIES

Under EASA AIR-OPS (CAT.OP.MPA.110), aerodrome operating minima establish legal limits for take-off and landing:

ILS Approach Category	Decision Height (DH)	Runway Visual Range (RVR) Minimum
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Category I (CAT I)	DH >= 200 ft (60 m)	RVR >= 550 meters (or 800 m for manual landing without HUD).
Category II (CAT II)	DH: 100 ft to < 200 ft (30 to 60 m)	RVR >= 300 meters (requires autoland or HUD guidance).
Category IIIa (CAT IIIa)	DH: < 100 ft (or NO DH)	RVR >= 175 meters (requires fail-passive autoland).
Category IIIb (CAT IIIb)	DH: < 50 ft (or NO DH)	RVR: 75 to < 175 meters (requires fail-operational autoland and rollout).
Category IIIc (CAT IIIc)	NO Decision Height (0 ft)	NO RVR limitation (Zero visibility landing and rollout - not operational).

2. LOW VISIBILITY TAKE-OFF (LVTO)

A take-off conducted with RVR < 400 meters is an LVTO operation requiring Specific Approval (Part-SPA.LVO). Absolute minimum RVR for commercial jets: 125 meters with runway centerline lights and markings (or 150 m for Cat C/D).

3. THE APPROACH BAN RULE

The flight crew may begin an instrument approach regardless of reported RVR. However:

- The 1,000 ft Ban Point: If the reported RVR is LESS than the published minimum, THE APPROACH MUST NOT BE CONTINUED BEYOND: The Outer Marker (OM), or an altitude of 1,000 ft above aerodrome elevation in the final approach segment!
- Ban Exemption After 1,000 ft: If the aircraft is ALREADY BELOW 1,000 ft AGL and RVR deteriorates below minima, the crew MAY CONTINUE to the Decision Height (DH). If visual references are acquired at DH, landing is permitted!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): The Approach Ban Decision Tree

Step 1: Check threshold point: 1,000 ft above aerodrome level (AAL) or Final Approach Fix (FAF). Step 2: Prior to reaching 1,000 ft AAL: - If reported RVR < published minimum -> APPROACH BAN! Cannot continue past 1,000 ft; execute go-around. - If reported RVR >= published minimum -> Cleared to continue past 1,000 ft AAL. Step 3: If below 1,000 ft AAL and reported RVR subsequently drops below minimum: - Legal permission to continue to Decision Altitude (DA) or MDA! Step 4: At DA/MDA: If required visual references are acquired -> LAND. If not acquired -> GO AROUND!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what are the Decision Height (DH) and minimum RVR limits for an ILS CATEGORY II approach?

- [A]: DH $\geq$ 200 ft, RVR $\geq$ 550 m
- [B]: DH between 100 ft and $<$ 200 ft, and RVR NOT LESS THAN 300 METERS
- [C]: DH $<$ 50 ft, RVR 75 m
- [D]: Zero DH

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cat II precision approach is defined by a DH between 100 and 199 ft, and a minimum RVR of 300 m.

Q2: According to the 'APPROACH BAN' rule, past which point is an aircraft prohibited from continuing an approach if the RVR drops below legal minima?

- [A]: Top of descent
- [B]: The 1,000 FT AGL POINT (or Outer Marker) in the final approach segment
- [C]: Initial approach fix
- [D]: Touchdown

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If RVR drops below legal minima before passing 1,000 ft AGL (or the outer marker), continuing the approach is illegal; a missed approach must be initiated.

Q3: What special equipment is required on the runway for a commercial jet Low Visibility Take-Off (LVTO) with a minimum RVR of 400 m?

- [A]: Only edge lights
- [B]: High-intensity RUNWAY CENTERLINE LIGHTS and runway markings
- [C]: Sodium flares
- [D]: Ground radar only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. LVTO below 400 m requires illuminated runway centerline lights and markings to provide continuous lateral visual tracking.